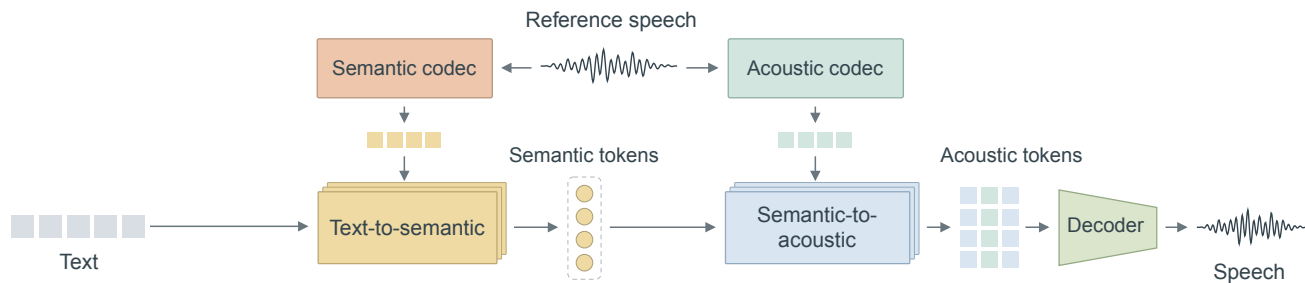


MaskGCT

Two-stage speech synthesis

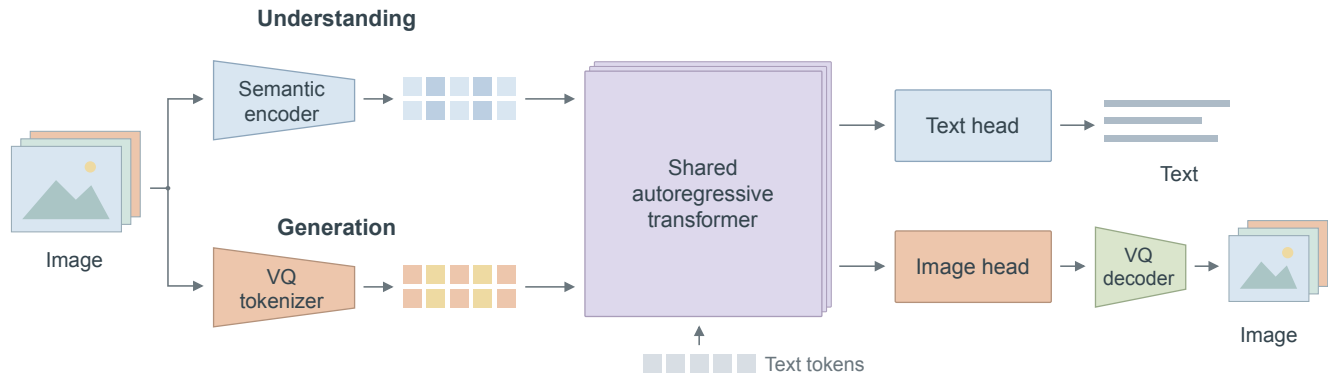


Reference speech conditions both semantic and acoustic generation. Two masked-prediction stages turn text into semantic tokens and then acoustic codes; the acoustic decoder reconstructs speech.

Inference topology only. Duration prediction, iterative refinement steps and the RVQ layer schedule are omitted.

Janus

Separate visual encoders. One shared model.

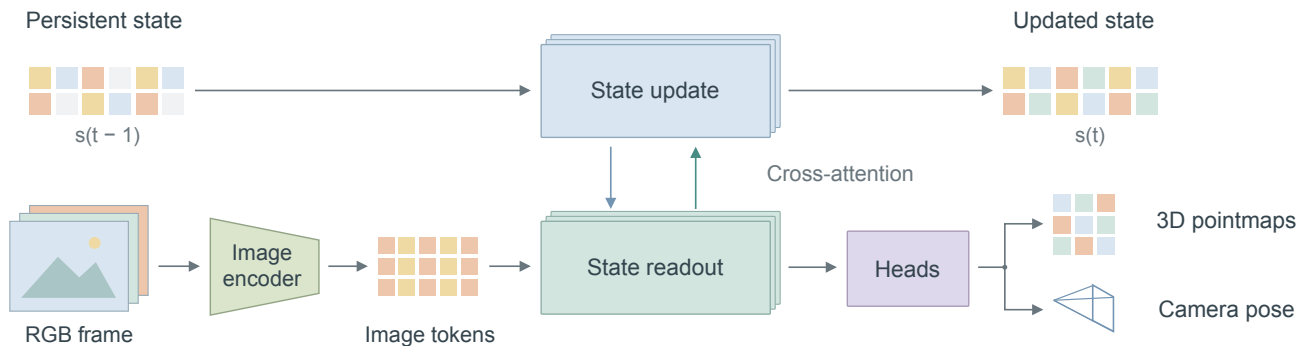


Task-specific encoders map semantic image features and discrete visual IDs to a shared autoregressive transformer. Separate heads predict text or image codes; a VQ decoder reconstructs images.

The two visual routes serve different tasks, not simultaneous inputs. The generation encoder supplies training image codes; text-to-image inference starts from text. Modality adapters are collapsed.

CUT3R

A persistent state that reads and updates

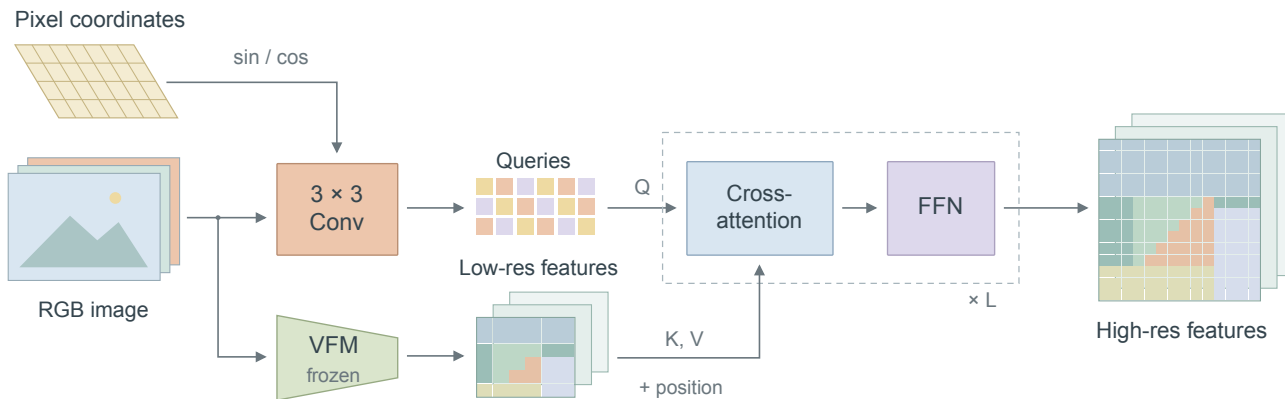


State and image streams interact through paired transformer decoders. The image stream updates the persistent state while reading accumulated context for pointmaps and camera pose.

One time step is shown. The pose token and individual output heads are abstracted; interleaved decoder interactions are condensed. Optional virtual-view queries are not shown.

LoftUp

Coordinate queries, high-resolution features



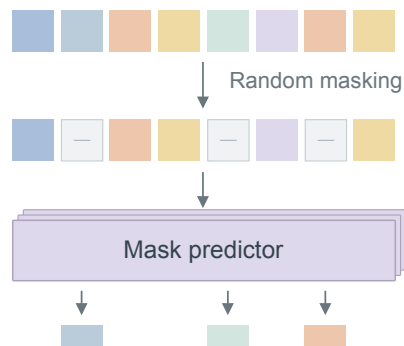
Pixel coordinates with sinusoidal encodings and RGB values form high-resolution queries. Low-resolution VFM features supply keys and values. Repeated cross-attention and feed-forward blocks predict high-resolution features.

Architecture only; the two-stage training scheme is omitted. Feature colors are illustrative tensors, not measured activations or qualitative model results.

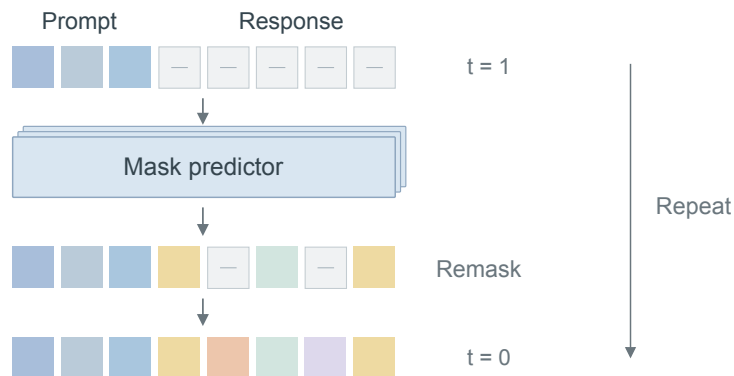
LLaDA

Masked prediction as a language model

(a) Training



(b) Parallel sampling

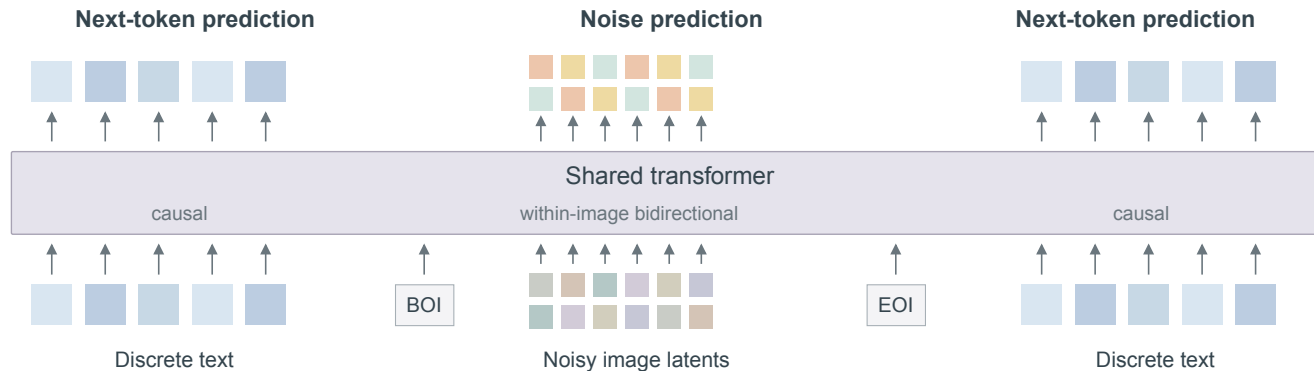


Training randomly masks tokens and supervises their recovery. During sampling, the prompt remains fixed while response masks are predicted in parallel; low-confidence predictions can be remasked.

Gray outlined cells denote masks. Only representative intermediate states are shown, not a fixed three-step sampler or an empirical decoding trace. SFT is omitted.

Transfusion

Discrete text meets continuous image latents



One transformer handles discrete text and continuous image latents. Text uses next-token prediction; images use diffusion noise prediction. Causal sequence attention is combined with bidirectional attention inside each image.

Modalities are grouped schematically, not drawn as exact shifted text targets. BOI/EOI mark image boundaries. VAE and modality-specific adapters are collapsed.